

Stokes' Theorem

Goal:

To evaluate a line integral w.r.t. a vector field using Stokes' Theorem.

The Curl of a Vector Field, $\mathbf{F}$, at a Point

Stokes' Theorem: _____ ,

Where S is an oriented piecewise smooth surface bounded by C , a piecewise smooth, positively oriented closed curve. Also, $\mathbf{F}$ has continuous first partials in an open region of space containing S .

Notes:

1. _____

2. Green's Theorem is a special case of Stokes' Theorem with _____

3. If $\text{curl } \mathbf{F} = \mathbf{0}$ throughout the domain of $\mathbf{F}$, then $\mathbf{F}$ is called _____

4. For a given boundary C of S , _____

Example:

Let $\mathbf{F}(x, y, z) = y\mathbf{i} + z\mathbf{j} - xy\mathbf{k}$ and S be oriented upward and given by $z = 4 - x - 2y$ in the first octant. Also, let C be the curve created when S intersects the coordinate planes. Find $\int_C \mathbf{F} \cdot d\mathbf{r}$.